FPGA Session Log — `final_8`: Hardware Bring-Up & Debug

Digilent Arty A7-35T · Open-Source Toolchain

Lucas Zhou

July 16–17, 2026

1 Session Goal

Build, flash, and verify the finished `top.v` (2-bit $\times$ 2-bit multiplier shown on a Pmod SSD seven-segment display) against the 2021 ECE 15A final, problem 8 — and debug why the display showed wrong or partial digits on real hardware.

2 Simulation (passed, unchanged all session)

The Icarus Verilog sweep of all 16 input combinations matches the case table in `top.v` exactly: products 0–9 as expected ($A \times B$ for $A, B \in \{0..3\}$). The logic was correct from the start; every bug below was a pin-mapping issue in `arty.xdc`.

```
source tools/oss-cad-suite/environment
iverilog -o tb_top.vvp tb_top.v top.v
vvp tb_top.vvp # 16/16 rows match the case table
```

3 Debug Timeline

1. **Flash #1 (XDC v1, seg[N] → AA..AG alphabetical):** scrambled shapes (a "7"-like glyph, fragments). Cause: `top.v` numbers segments *geometrically bottom-to-top* (seg[0..6] = D,C,E,G,B,F,A), not alphabetically.
2. **Flash #2 (XDC v2, permuted, still JA-only):** input `1011` (2×3) showed only *part* of a 6.
3. **Photo analysis (`IMG_8616`):** two findings — (a) the Pmod SSD straddles *two* connectors: header J1 (AA–AD) in JA's top row, header J2 (AE, AF, AG, CAT) in *JB's* top row, so v1/v2 drove three segments + digit select on untouched JA bottom-row pins (they floated); (b) the glyphs are mounted 180° rotated (decimal points top-left), so each signal lights the segment opposite its name.
4. **Flash #3 (XDC v3, current):** AE/AF/AG/CAT moved to JB pins E15/E16/D15/C15; all seven assignments rotated 180°. Digit select `C` now actually driven by `assign C = 1'b1`.

4 Final Pin Map (XDC v3)

Port	Meaning	SSD signal	Pin	Connector
`A2/A1/B2/B1`	inputs	—	A10/C10/C11/A8	SW3/SW2/SW1/SW0
`seg[0]`	bottom	AA	G13	JA1
`seg[1]`	lower-right	AF	E16	JB2
`seg[2]`	lower-left	AB	B11	JA2
`seg[3]`	middle	AG	D15	JB3
`seg[4]`	upper-right	AE	E15	JB1
`seg[5]`	upper-left	AC	A11	JA3
`seg[6]`	top	AD	D12	JA4
`C`	digit select	CAT	C15	JB4

5 Open Items

- **Visual confirmation** of XDC v3 on hardware (expect a complete upright 6 for input `1011`; spot-check 9, 4, 1, 0).
- If the *other* digit of the two should light: flip `assign C = 1'b1` to `1'b0` in `top.v`, rebuild.
- Current load is **SRAM only** (lost on power-off). For the final demo, write to flash: `openFPGALoader -b arty -f top.bit`.

6 Files in `projects/final_8/`

File	Status	Notes
`top.v`	unchanged, correct	author's multiplier + case table
`arty.xdc`	rewritten twice (v3)	JA/JB split + 180° rotation
`xdc-explained.pdf`	new this session	line-by-line XDC tutorial + full debug story
`session-log-2026-07-13.pdf`	prior session	syntax fixes, testbench, sim flow
`top.bit`	flashed to SRAM	3 builds, all 0 errors

For the beginner-level, line-by-line explanation of the XDC syntax and all three versions, see `xdc-explained.pdf`.